

Long-range order and spontaneous symmetry breaking in quantum spin systems and hard-core bosons on a lattice

***part 4 LRO and SSB in the ground states of
the quantum XY model***

***Advanced Topics in
Statistical Physics
by Hal Tasaki***

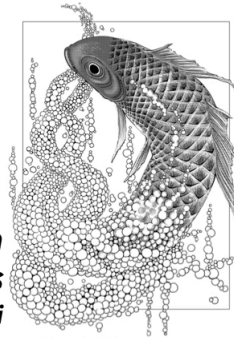


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§ Hamiltonian and order operators

XY model (ferromagnetic, but see p.8)

$$(\hat{S}_u^\pm = \hat{S}_u^{(x)} \pm i \hat{S}_u^{(y)})$$

$$\hat{H}_L = - \sum_{\{u,v\} \in \mathcal{B}_L} (\hat{S}_u^{(x)} \hat{S}_v^{(x)} + \hat{S}_u^{(y)} \hat{S}_v^{(y)}) = - \frac{1}{2} \sum_{\{u,v\} \in \mathcal{B}_L} (\hat{S}_u^+ \hat{S}_v^- + \hat{S}_u^- \hat{S}_v^+) \quad (1)$$

symmetry

$$(\hat{T}_w : \text{translation} \quad \hat{T}_w^\dagger \hat{H}_L \hat{T}_w = \hat{H}_L \quad \text{for any } w \quad (2))$$

$$[\hat{H}_L, \hat{O}_L^{(z)}] = 0 \quad (3)$$

$$(\hat{U}_L^{(z,\theta)})^\dagger \hat{H}_L \hat{U}_L^{(z,\theta)} = \hat{H}_L \quad (4) \quad \text{for any } \theta \rightarrow \begin{matrix} \text{continuous} \\ \text{U(1) symmetry} \end{matrix}$$

order operators $(\hat{O}_L^{(x)}, \hat{O}_L^{(y)})$ $\hookrightarrow$ magnetization vector in the xy plane

see part 1 - p7, (3), (4) for the transformation properties

$$[\hat{H}_L, \hat{O}_L^{(\alpha)}] \neq 0 \quad \text{for } \alpha = x, y \quad (5)$$

order operators and Hamiltonian do not commute!!

§ properties of the exact ground state

uniqueness and symmetry

theorem (Mattis 1979) $\hat{H}_L$ has a unique ground state $|\Phi_L^{GS}\rangle$

we further have $\hat{O}_L^{(z)} |\Phi_L^{GS}\rangle = 0$ (1)

and hence $\hat{U}_L^{(z, \theta)} |\Phi_L^{GS}\rangle = |\Phi_L^{GS}\rangle$ for any θ (2)

we also have $\hat{T}_w |\Phi_L^{GS}\rangle = |\Phi_L^{GS}\rangle$ for any $w \in \mathbb{Z}^d$ (3)

(proof: Perron-Frobenius type. not too difficult. (see, e.g., my book))

$$(2) \rightarrow \langle \Phi_L^{GS} | \hat{O}_L^{(x)} | \Phi_L^{GS} \rangle = \langle \Phi_L^{GS} | \hat{O}_L^{(y)} | \Phi_L^{GS} \rangle = 0 \quad (4)$$

no symmetry breaking (as expected)

E_L^{GS} : the ground state energy

$$\epsilon^{GS} = \lim_{L \uparrow \infty} \frac{1}{L^d} E_L^{GS} \quad (5) \quad \text{the ground state energy density}$$

the existence of long-range order

Theorem (Kennedy, Lieb, Shastri 1988, Kubo, Kishi 1988)

for $d=2,3,\dots$, there is a (d -dependent) constant $\varrho_0 > 0$ st.

$$\langle \Phi_L^{GS} | \left(\frac{\hat{O}^{(x)}}{L^d} \right)^2 | \Phi_L^{GS} \rangle = \langle \Phi_L^{GS} | \left(\frac{\hat{O}^{(y)}}{L^d} \right)^2 | \Phi_L^{GS} \rangle \geq \varrho_0 \quad (1) \text{ for any } L$$

this suggests $\langle \Phi_L^{GS} | \hat{S}_u^{(\alpha)} \hat{S}_v^{(\alpha)} | \Phi_L^{GS} \rangle \simeq \varrho^* \geq \varrho_0$ for large $|u-v|$ (2)

$\alpha = x, y$

LRO in the xy plane

we again see

$$\sqrt{\langle \Phi_L^{GS} | \left(\frac{\hat{O}^{(\alpha)}}{L^d} \right)^2 | \Phi_L^{GS} \rangle - \left(\langle \Phi_L^{GS} | \frac{\hat{O}^{(\alpha)}}{L^d} | \Phi_L^{GS} \rangle \right)^2} \geq \sqrt{\varrho_0} \text{ for } \alpha = x, y \quad (3)$$

unphysical!

fluctuation of $\hat{O}^{(\alpha)} / L^d$ in $|\Phi_L^{GS}\rangle$

proof: relies on the method of reflection positivity $\rightarrow$ highly nontrivial

equilibrium for $\begin{cases} \text{classical} & \text{Fröhlich, Simon, Spencer 1978} \\ \text{quantum} & \text{Dyson, Lieb, Simon 1978} \end{cases}$

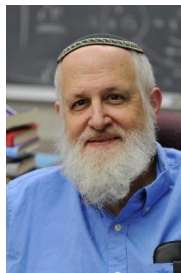
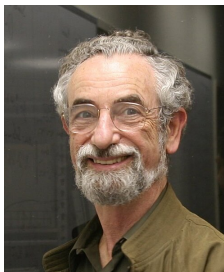
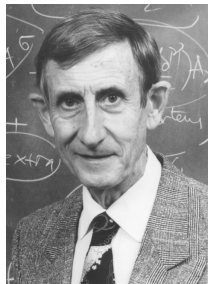
Phase Transitions in Quantum Spin Systems with Isotropic and Nonisotropic Interactions

Freeman J. Dyson,¹ Elliott H. Lieb,² and Barry Simon^{2,3}

Received November 1, 1976

We prove the existence of spontaneous magnetization at sufficiently low temperature, and hence of a phase transition, in a variety of quantum spin systems in three or more dimensions. The isotropic spin $1/2$ x - y model and the Heisenberg antiferromagnet with spin $1, 3/2, \dots$ and with nearest neighbor interactions on a simple cubic lattice are included.

KEY WORDS: Phase transitions; Heisenberg ferromagnet.



XY model

$$\hat{H}_L = - \sum_{\{u,v\} \in \mathcal{B}_L} (\hat{S}_u^{(x)} \hat{S}_v^{(x)} + \hat{S}_u^{(y)} \hat{S}_v^{(y)}) \quad (1)$$

unphysical!

unique exact ground state $|\Phi_L^{GS}\rangle$ exhibits LRO but no SSB in $d \geq 2$

there must be physical macroscopic ground states
with LRO and SSB

$U(1)$ symmetry is broken $\rightarrow$ infinitely many macroscopic ground states

$\downarrow$
ever-increasing number of low-lying states

Anderson's
tower

we need an extension of the Horsch-vonder Linden theory to models
with a continuous symmetry that is spontaneously broken!

done in Koma-Tasaki 1994

as a spin-off of Koma-Tasaki 1993 $\rightarrow$

Griffiths theorem for the antiferromagnetic Heisenberg model

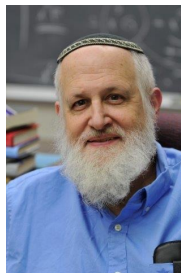
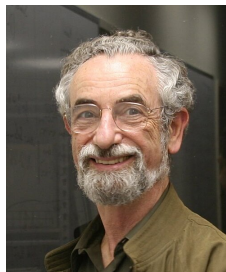
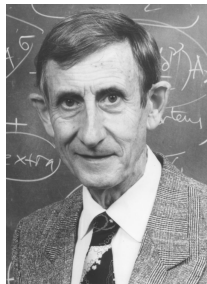
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Following Griffiths,⁽¹⁶⁾ we can apply Corollary 1.2 to prove that long-range order in the sense of (4) implies a spontaneous magnetization. Let H_Λ be the Hamiltonian of a system in a box with periodic boundary conditions (but with no restriction on the form of the interaction). Let A_α be an operator at site α . Define $m(A)$ by

$$m(A) = \lim_{\mu \downarrow 0} \frac{d}{d\mu} \left(\lim_{|\Lambda| \rightarrow \infty} |\Lambda|^{-1} \ln \left\{ \text{Tr} \left[\exp \left(-\beta H_\Lambda + \mu \sum A_\alpha \right) \right] \right\} \right)$$

As an immediate consequence of Corollary 1.2, we have the following result:

Theorem 1.2. Under the above conditions, if $\sum_{\alpha \in \Lambda} A_\alpha$ commutes with H_Λ , then

$$m(A) \geq \overline{\lim}_{|\Lambda| \rightarrow \infty} \left\langle \left(|\Lambda|^{-1} \sum_{\alpha \in \Lambda} A_\alpha \right)^{2k} \right\rangle_\Lambda^{1/2k} \quad (14)$$

Remark 2. If $\text{Tr}[\exp(-\beta H_\Lambda + \mu \sum A_\alpha)]$ is replaced by $\text{Tr}[\exp(-\beta H_\Lambda) \exp(\mu \sum A_\alpha)]$, the commutation condition is not needed; the commutation condition is only used to obtain the physically relevant object. The commutativity unfortunately fails in several cases of interest, notably the x - y and the antiferromagnetic models. In Section 5, after Theorem 5.2, we develop a different strategy for proving the existence of a phase transition in the noncommutative case, and apply it there to the x - y model. In Section 6 we use it again for the antiferromagnet.



Tohru Koma

Oct. 1988



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Symmetry Breaking in Heisenberg Antiferromagnets

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Abstract: We extend Griffith's theorem on symmetry breaking in quantum spin systems to the situation where the order operator and the Hamiltonian do not commute with each other. The theorem establishes that the existence of a long range order in a symmetric (non-pure) infinite-volume state implies the existence of a symmetry breaking in the state obtained by applying an infinitesimal symmetry-breaking field. The theorem is most meaningful when applied to a class of quantum antiferromagnets where the existence of a long range order has been proved by the Dyson-Lieb-Simon method. We also present a related theorem for the ground states. It is an improvement of the theorem by Kaplan, Horsch and von der Linden. Our lower bounds on the spontaneous staggered magnetization in terms of the long range order parameter take into account the symmetry of the system properly, and are likely to be saturated in general models.

technical remark

The theorems mentioned above are for the antiferromagnetic model

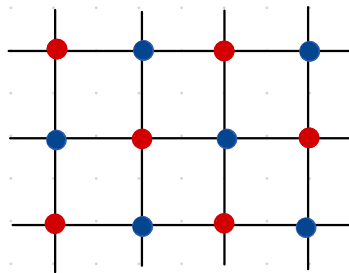
$$\hat{H}_L = \sum_{\{u,v\} \in \mathcal{B}_L} (\hat{S}_u^{(x)} \hat{S}_v^{(x)} + \hat{S}_u^{(y)} \hat{S}_v^{(y)} + \lambda \hat{S}_u^{(z)} \hat{S}_v^{(z)}) \quad (1) \quad \text{with some } \lambda \in [0,1]$$

Λ_L is bipartite $\Lambda_L = A \cup B, A \cap B = \emptyset$ (2)

$$\{u,v\} \in \mathcal{B}_L \Rightarrow u \in A, v \in B \text{ or } u \in B, v \in A \quad (3)$$

if we define

$$\hat{V}_L = \bigotimes_{u \in A} e^{-i\pi \hat{S}_u^{(z)}} \quad (4)$$



$$\hat{V}_L^\dagger \hat{H}_L \hat{V}_L = \sum_{\{u,v\} \in \mathcal{B}_L} (-\hat{S}_u^{(x)} \hat{S}_v^{(x)} - \hat{S}_u^{(y)} \hat{S}_v^{(y)} + \lambda \hat{S}_u^{(z)} \hat{S}_v^{(z)}) \quad (5)$$

ferromagnetic for $\lambda=0$

low-lying states

9

$$\hat{O}_L^{\pm} = \hat{O}_L^{(x)} \pm i \hat{O}_L^{(y)} \quad (1) \quad \text{spin raising and lowering operators}$$

$$[\hat{O}_L^{(z)}, \hat{O}_L^{\pm}] = \pm \hat{O}_L^{\pm} \quad (2)$$

for $M=1, 2, \dots$

$$\left\{ \begin{array}{l} |\Gamma_L^M\rangle = \frac{(\hat{O}_L^+)^M |\Phi_L^{GS}\rangle}{\|(\hat{O}_L^+)^M |\Phi_L^{GS}\rangle\|} \end{array} \right. \quad (4)$$

$$\left\{ \begin{array}{l} |\Gamma_L^{-M}\rangle = \frac{(\hat{O}_L^-)^M |\Phi_L^{GS}\rangle}{\|(\hat{O}_L^-)^M |\Phi_L^{GS}\rangle\|} \end{array} \right. \quad (5)$$

we set $|\Gamma_L^0\rangle = |\Phi_L^{GS}\rangle$

$$\text{note } \hat{O}_L^{(z)} |\Gamma_L^M\rangle = M |\Gamma_L^M\rangle \quad (6)$$

for $M=0, \pm 1, \pm 2, \dots$

$$\text{thus } \langle \Gamma_L^M | \Gamma_L^{M'} \rangle = \delta_{M, M'} \quad (7)$$

Theorem (Koma, Tasaki 1994)

for $d=2,3,\dots$ there are (d -dependent) constants C_1, C_2 s.t.

$$0 \leq \langle \Gamma_L^M | \hat{H}_L | \Gamma_L^M \rangle - E_L^{\text{gs}} \leq C_1 \frac{M^2}{L^d} \quad (1)$$

for any M with $|M| \leq C_2 L^{d/2}$ and any L → expected scaling

(proof: a patient recursive version of the double commutator estimate as in Horsch, von der Linden)
ever increasing number of low-lying states as L grows!

note $\hat{T}_W |\Gamma_L^M\rangle = |\Gamma_L^M\rangle$ (2) $\lim_{L \rightarrow \infty} \frac{1}{L^d} \langle \Gamma_L^M | \hat{H}_L | \Gamma_L^M \rangle = E_{\text{gs}}$ (3)

$|\Gamma_L^M\rangle$ with $|M| \leq C_2 L^{d/2}$ is a macroscopic ground state → $\langle \Psi_L^M | \Psi_L^{M'} \rangle = \delta_{M,M'}$

corollary for $d=2,3,\dots$, there is an energy eigenstate $|\Psi_L^M\rangle$ with eigenvalue E_L^M s.t.

$$0 \leq E_L^M - E_L^{\text{gs}} \leq C_1 \frac{M^2}{L^d} \quad (4)$$

for any M with $|M| \leq C_2 L^{d/2}$ and any L

many low-lying energy eigenstates!

Anderson's tower

Anderson's tower of states Anderson 1952,

11

- $\hat{H}_L$ has a continuous symmetry
- $[\hat{H}_L, \text{order operator}] \neq 0$
- the ground state is in the ordered phase

in many cases



unique ground state with LRO but without SSB



"tower" of low-lying energy eigenstates



Philip Anderson

The earliest clear exact-diagonalization observation of the "tower" structure (that I know of)
2D antiferromagnetic Heisenberg model

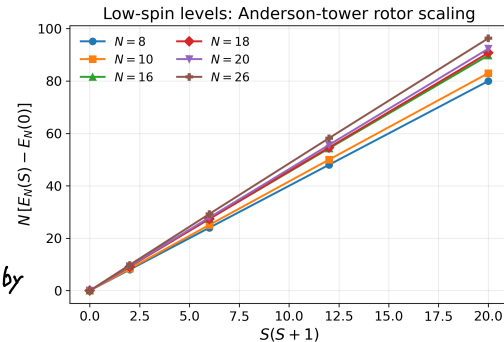
Table I. The lowest energies for all the possible total spin S .

S	$N=8$	10	16	18	20	26
0	-6.00000	-7.30007	-11.22848	-12.49175	-13.81616	-17.79576
1	-5.00000	-6.46410	-10.64988	-11.98334	-13.34919	-17.41911
2	-3.00000	-4.79129	-9.51769	-10.97147	-12.42084	-16.67010
3	0.00000	-2.30278	-7.83654	-9.45968	-11.03610	-15.55848
4	4.00000	1.00000	-5.61728	-7.44944	-9.20045	-14.09109
5		5.00000	-2.86620	-4.96548	-6.92616	-12.27393
6			0.35425	-2.02379	-4.23611	-10.11683
7			4.00000	1.31217	-1.15569	-7.63527
8			8.00000	5.00000	2.26795	-4.84547
9				9.00000	6.00000	-1.76397
10					10.00000	1.58929
11						5.18590
12						9.00000
13						13.00000

Reproduced with permission from Macoto Kikuchi, Yutaka Okabe, and Seiji Miyashita, "On the Ground-State Phase Transition of the Spin 1/2 XXZ Model on the Square Lattice", J. Phys. Soc. Jpn. 59, 492-496 (1990).

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$$E_S - E_0 \simeq \text{const.} \frac{S(S+1)}{N} \quad \left(\begin{array}{l} S \leftrightarrow M \\ N \leftrightarrow L^d \end{array} \right)$$



Data redrawn from Table I of Kikuchi, Okabe, and Miyashita, J. Phys. Soc. Jpn. 59, 492-496 (1990).

§ macroscopic ground states with full symmetry breaking

12

▷ symmetry breaking order parameter

$$m^* = \lim_{b \uparrow \infty} \lim_{L \uparrow \infty} \left\{ \langle \Phi_L^{GS} | \left(\frac{\hat{O}_L^{(\alpha)}}{L^d} \right)^{2b} | \Phi_L^{GS} \rangle \right\}^{\frac{1}{2b}} \quad (1) \quad \alpha = x, y$$

→ the "maximum possible value" of $\hat{O}^{(\alpha)}/L^d$ in the ground state (as $L \uparrow \infty$)

Theorem (Koma, Tasaki 1993, 1994)

$$m^* \geq \sqrt{2q_0} \quad (2)$$

heuristic picture:

the order operator $(\hat{O}_L^{(x)}/L^d, \hat{O}_L^{(y)}/L^d)$ is essentially
a two-dimensional vector $\vec{m} = (m_x, m_y)$ s.t. $|\vec{m}| = m^*$

$$\text{then } m_x^2 + m_y^2 = (m^*)^2 \rightarrow (m^*)^2 \simeq \langle m_x^2 \rangle + \langle m_y^2 \rangle \geq 2q_0$$

(we proved $m^* \geq \sqrt{3q_0}$ for the antiferromagnetic Heisenberg model)

► Symmetry breaking states

$M(L)$ a function that diverges as $L \uparrow \infty$ not too rapidly

$$|\square_L^0\rangle = \frac{1}{\sqrt{2M(L)+1}} \sum_{M=-M(L)}^{M(L)} |\square_L^M\rangle = \frac{1}{\sqrt{2M(L)+1}} \sum_{M=-M(L)}^{M(L)} \frac{(\hat{O}_L^+)^M |\Phi_L^{\text{GS}}\rangle}{\|(\hat{O}_L^+)^M |\Phi_L^{\text{GS}}\rangle\|} \quad (1)$$

$$|\square_L^\theta\rangle = \hat{U}_L^{(z,\theta)} |\square_L^0\rangle = \frac{1}{\sqrt{2M(L)+1}} \sum_{M=-M(L)}^{M(L)} e^{-i\theta M} |\square_L^M\rangle \quad (2) \quad \uparrow \left((\hat{O}_L^+)^M = (\hat{O}_L^-)^{|M|} \text{ for } M < 0 \right)$$

$$\hat{T}_W |\square_L^\theta\rangle = |\square_L^\theta\rangle \quad (3) \quad \lim_{L \uparrow \infty} \frac{1}{L^d} \langle \square_L^\theta | \hat{H}_L | \square_L^\theta \rangle = E_{\text{GS}} \quad (4)$$

$|\square_L^0\rangle$ is a macroscopic ground state

Theorem (Koma, Tasaki 1994, Tasaki 2019)

$$\lim_{L \uparrow \infty} \langle \square_L^\theta | \frac{\hat{O}_L^{(\alpha)}}{L^d} | \square_L^\theta \rangle = \begin{cases} m^* \cos \theta, & \alpha = x \\ m^* \sin \theta, & \alpha = y \end{cases} \quad (5)$$

$$\lim_{L \uparrow \infty} \langle \square_L^\theta | \left(\frac{\hat{O}_L^{(\alpha)}}{L^d} \right)^2 | \square_L^\theta \rangle = \begin{cases} (m^* \cos \theta)^2, & \alpha = x \\ (m^* \sin \theta)^2, & \alpha = y \end{cases} \quad (6)$$

the 2019 theorem implies in particular

14

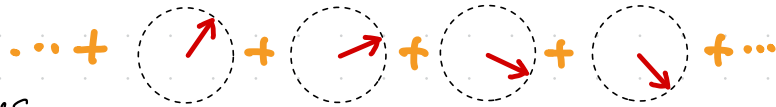
$$\lim_{L \rightarrow \infty} \sqrt{\langle \frac{\vec{\Gamma}}{L}^\theta | \left(\frac{\hat{\vec{O}}_L^{(\alpha)}}{L^d} \right)^2 | \frac{\vec{\Gamma}}{L}^\theta \rangle - \left(\langle \frac{\vec{\Gamma}}{L}^\theta | \frac{\hat{\vec{O}}_L^{(\alpha)}}{L^d} | \frac{\vec{\Gamma}}{L}^\theta \rangle \right)^2} = 0 \quad (1) \quad \alpha = x, y, z$$

the order operators $(\hat{\vec{O}}_L^{(\alpha)}/L^d, \hat{\vec{O}}_L^{(\alpha)}/L^d)$ take definite values $(m^* \cos \theta, m^* \sin \theta)$

$|\frac{\vec{\Gamma}}{L}^\theta\rangle$ is a physical macroscopic ground state with full SSB

relation to the exact ground state

p13-(2), with $|\frac{\vec{\Gamma}}{L}^0\rangle = |\Phi_L^{GS}\rangle$, shows



$$|\Phi_L^{GS}\rangle = \frac{\sqrt{2M(L)+1}}{2\pi} \int_0^{2\pi} d\theta |\frac{\vec{\Gamma}}{L}^\theta\rangle \quad (2)$$

a remarkable Schrödinger's cat where physical macroscopic ground states with all $\theta \in [0, 2\pi]$ are superposed!!

it must collapse into $|\frac{\vec{\Gamma}}{L}^\theta\rangle$ with some θ



► the expectation value of a $U(1)$ invariant operator

$\hat{A}$: a local operator (any polynomial of $\hat{S}_u^{(\alpha)}$, $\alpha=x,y,z$, $u \in \mathbb{Z}^d$)

such that $[\hat{A}, \hat{O}_L^{(2)}] = 0$ (1)

|| $\rightarrow$ well-defined if we take sufficiently large L
 $\hat{A}$ is $U(1)$ invariant

then $\langle \Xi_L^\theta | \hat{A} | \Xi_L^\theta \rangle = \frac{1}{2M(L)+1} \sum_{M=-M(L)}^{M(L)} \langle \Gamma_L^M | \hat{A} | \Gamma_L^M \rangle$ (2)

independent of θ

conjecture for any $U(1)$ invariant local operator $\hat{A}$

$\lim_{L \uparrow \infty} |\langle \Xi_L^\theta | \hat{A} | \Xi_L^\theta \rangle - \langle \Phi_L^{\text{GS}} | \hat{A} | \Phi_L^{\text{GS}} \rangle| = 0$ (3)

(justified if the ground state has a proper clustering property)

▶ Griffiths-type theorem

16

The theorem by Kaplan, Horsch, and von der Linden is automatically generalized by using $|\Xi_L^0\rangle$ as the trial state

$$\hat{H}_L^h = \hat{H}_L - h \hat{O}_L^{(x)} \quad (1) \quad |\Phi_L^{GS,h}\rangle \text{ an exact ground state of } \hat{H}_L^h$$

 symmetry breaking field in the x-direction

$$\lim_{h \downarrow 0} \lim_{L \uparrow \infty} \langle \Phi_L^{GS,h} | \frac{\hat{O}_L^{(x)}}{L^d} | \Phi_L^{GS,h} \rangle \geq m^* \geq \sqrt{2g_0} > 0 \quad (2)$$

infinitesimal symmetry breaking field leads to a ground state with SSB